

MACHINE LEARNING FINAL PROJECT · JULY 2026

Ensemble Methods and Unsupervised Learning

When Does Boosting Win, and When Does Bagging Win?

PRESENTED BY

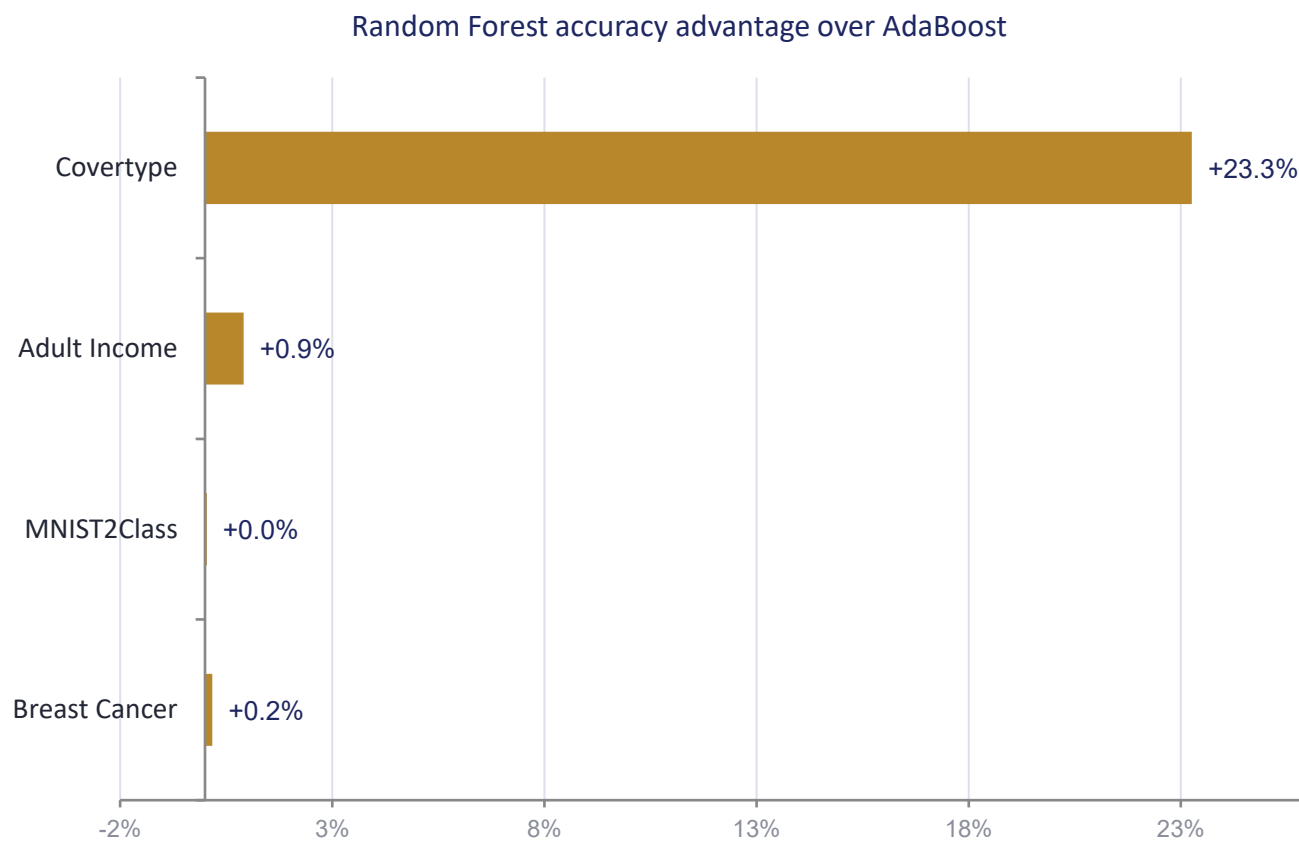
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THE QUESTION

When Does Boosting Win, and When Does Bagging Win?

Model choice depends on dataset separability, feature interactions, class imbalance, label noise, and the capacity of the weak learner — not on which algorithm is fashionable.



Near-tie

Breast Cancer & MNIST2Class: separable data, either model works (< 0.2 pt gap).

RF safer

Adult Income: mixed categorical/numeric features and interactions favor bagging (+0.9 pt, significant).

RF dominates

Covertypes: severe 7-class imbalance breaks SAMME boosting; RF wins by 23 points.

Four Datasets, Four Different Challenges

Dataset	Rows	Feature Type	Classes	Experimental Role
Breast Cancer Wisconsin	569	30 numeric	2	High-dimensional, low-sample baseline
Adult Income	48,842	6 numeric + 8 categorical	2	Mixed feature types, real-world interactions
Covertypes	50,000	54 numeric / binary	7	Severe class imbalance (largest:smallest $\approx$ 77:1)
MNIST2Class	14,780	784 pixel (numeric)	2	Very high-dimensional, visually separable

Covertypes's class-4 minority holds only 236 of 50,000 rows — the sharpest imbalance in the study.

A Leakage-Safe Experimental Pipeline



**Covertime only — applied identically to every compared model, within training folds only.*

- Numeric imputers / scalers and categorical encoders are fit on training rows only, then applied to held-out rows.
- Adult Income keeps all 48,842 rows; unseen categorical levels at test time map to all-zero indicator groups.
- Every compared model (Tree, AdaBoost, Random Forest, sklearn reference) sees identical 5-fold splits.
- One canonical seed — 42 — governs selection, splitting, cross-validation, corruption and model initialization.

What We Implemented, From Scratch

SUPERVISED (ENSEMBLES)

Decision Tree

Custom CART-style splitter; sklearn used only as a labelled reference baseline.

AdaBoost — SAMME / SAMME.R

Sequential reweighting of misclassified rows with staged-prefix tracking.

Random Forest

Bootstrap sampling + random feature subsets, OOB scoring, multiprocessing.

Gradient Boosting (bonus)

Simplified residual-fitting booster, benchmarked against AdaBoost.

UNSUPERVISED

PCA

Variance-preserving projection; scree curves drive component selection.

K-Means

Custom Lloyd's-algorithm implementation with elbow-based k selection.

DBSCAN

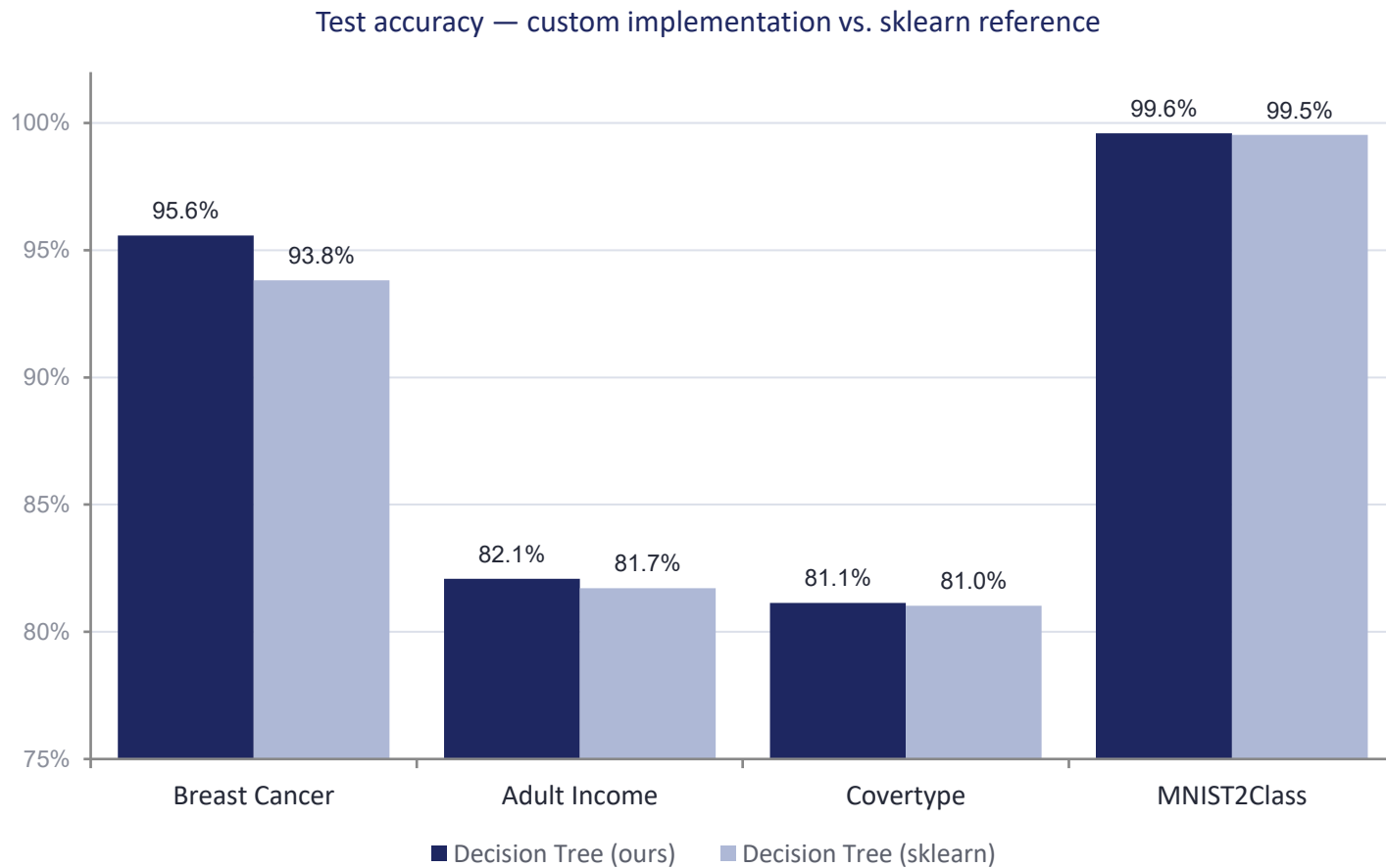
Exact KD-tree neighbor search for density-based clustering.

t-SNE (visualization)

sklearn utility, used strictly for 2-D visual inspection, not for metrics.

Baseline Validation: Custom Tree vs. sklearn

Before trusting ensemble results, the base learner itself must track its library reference.



1.77 pt

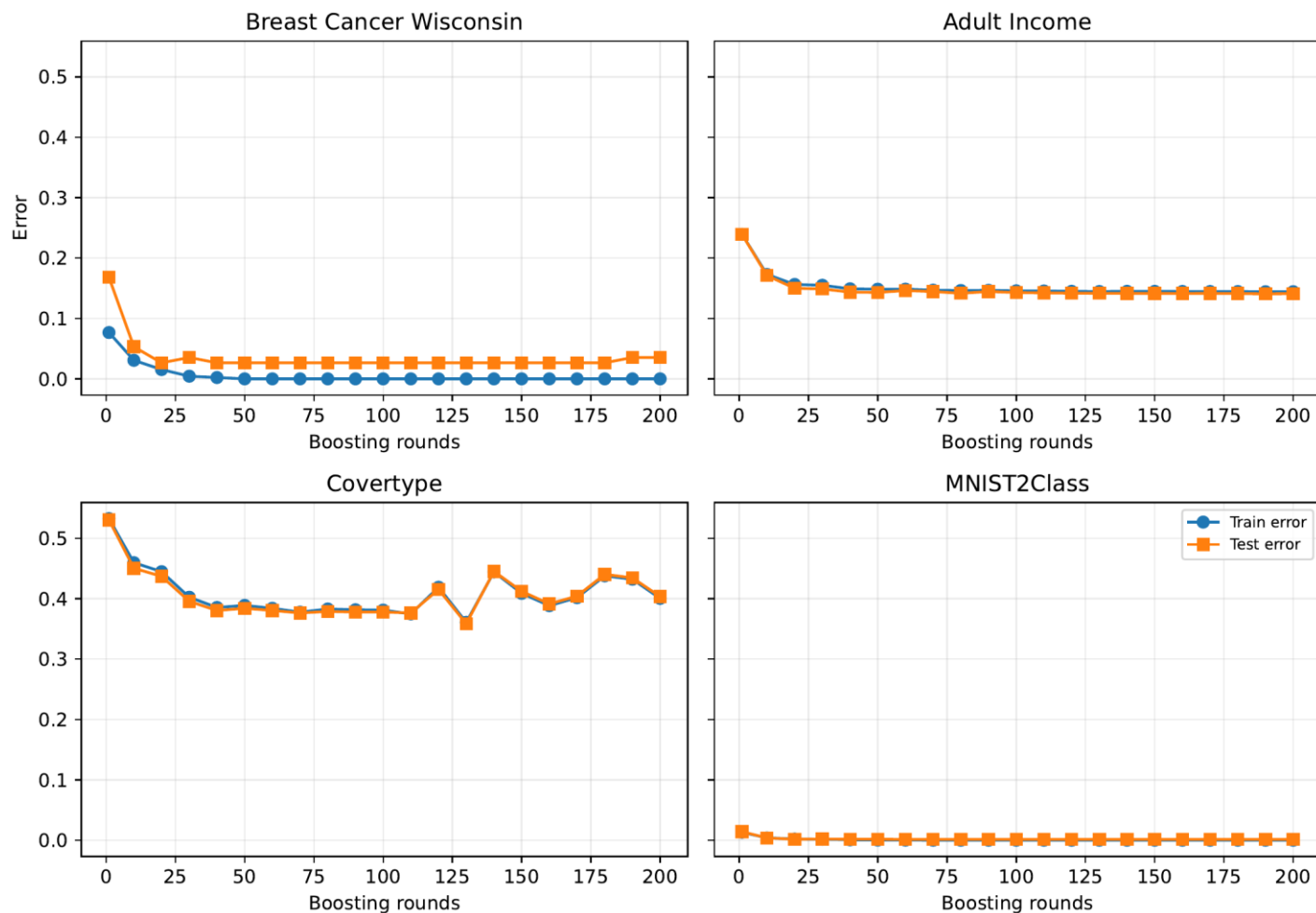
Largest gap — Breast Cancer Wisconsin

0.07 pt

Smallest gap — MNIST2Class

Every gap stays under 2 percentage points. This confirms our from-scratch splitter, impurity criterion and stopping rules behave credibly before we build AdaBoost and Random Forest on top of them.

AdaBoost: Sequential Correction at Scale



Test error vs. number of estimators, all four datasets (adaboost_scaling.csv)

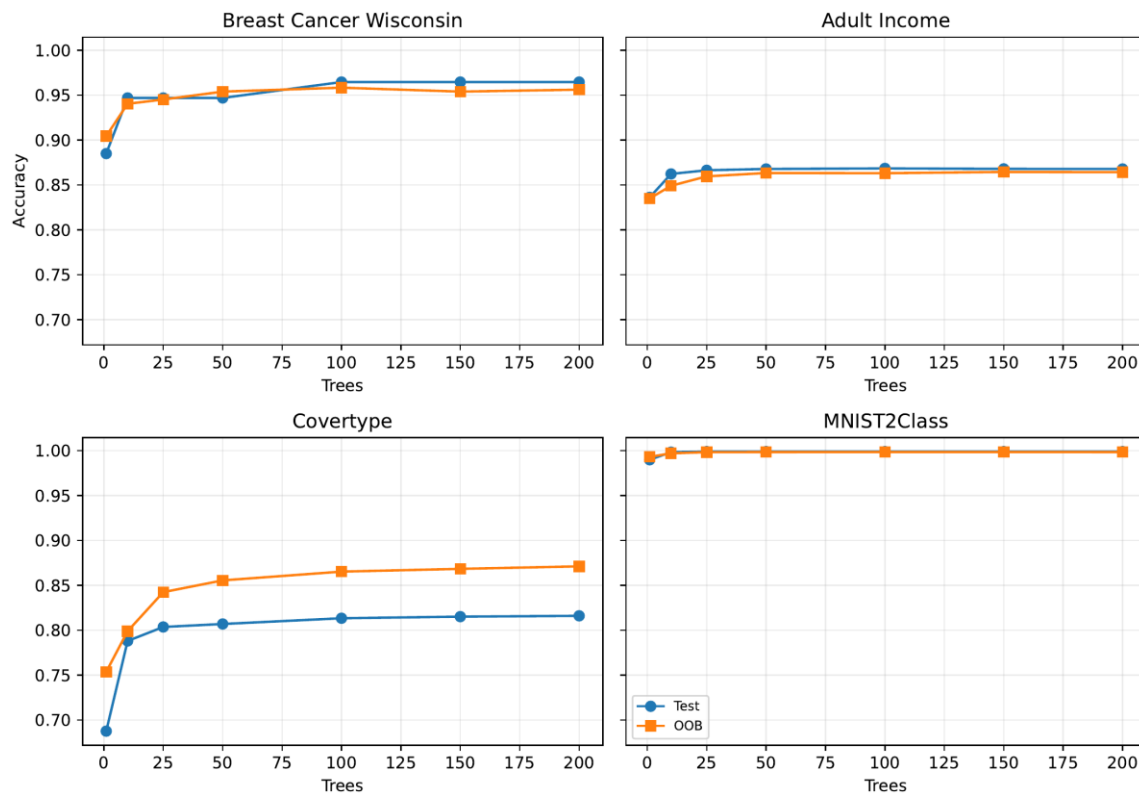
Each round reweights misclassified rows so the next weak learner focuses on them.

SAMME.R uses class-probability estimates for smoother, faster-converging updates.

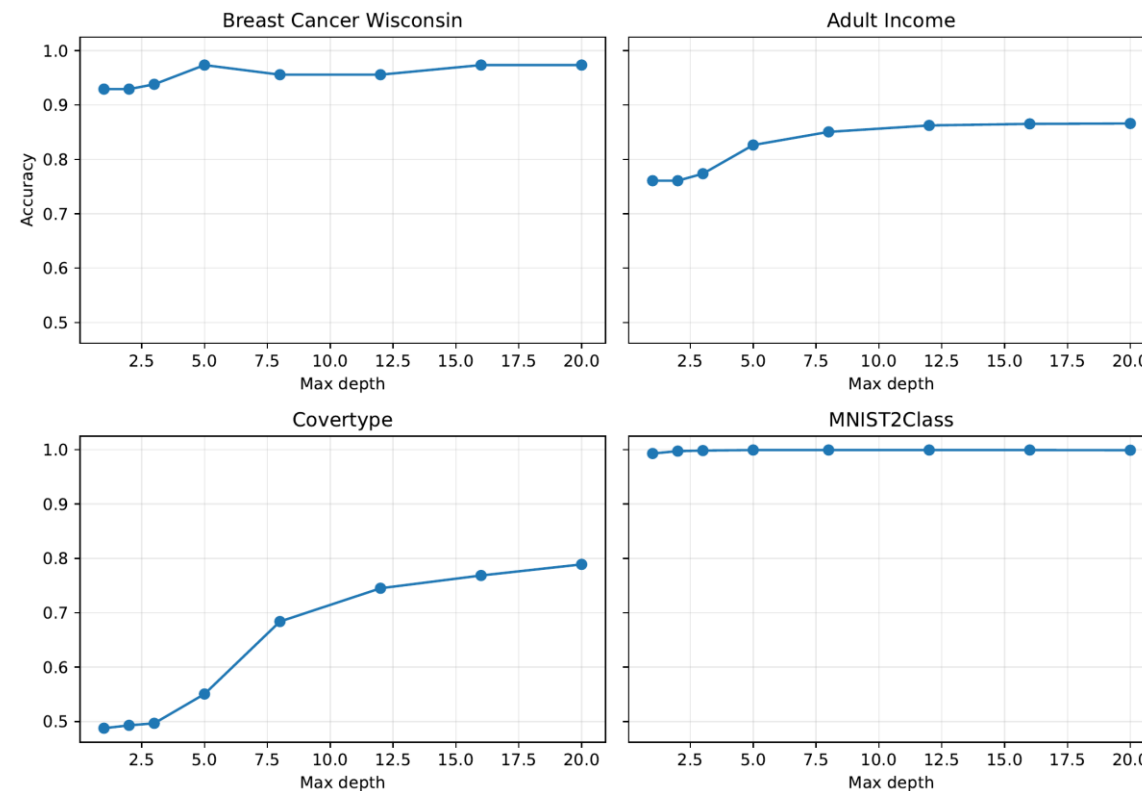
MNIST2Class saturates within ~30 estimators — the classes are already near-separable.

Covertypes error stays high and volatile — severe 7-class imbalance destabilizes SAMME's weighting.

Random Forest: Averaging Away Variance



Accuracy vs. tree count



Accuracy vs. max tree depth

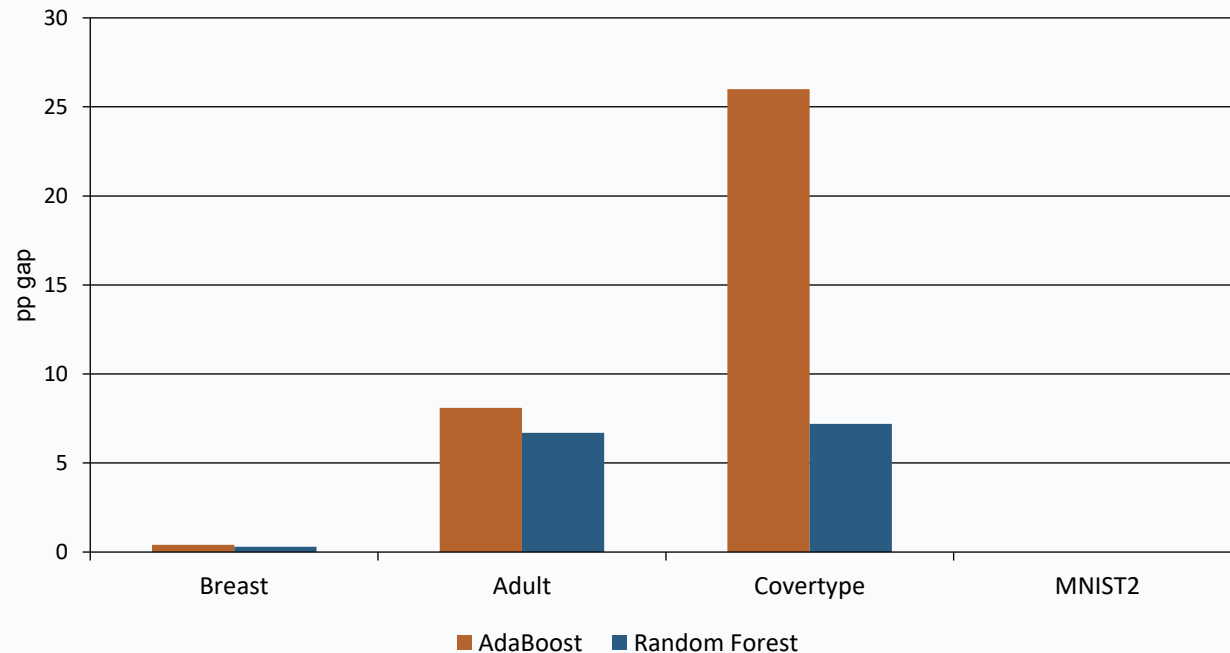
- Bootstrap resampling + random feature subsets decorrelate trees before hard-vote averaging.
- Out-of-bag (OOB) score tracks generalization without a held-out split, closely mirroring test accuracy.
- Gains are steep to ~25 trees, then diminishing — more trees mainly reduce variance, not bias.
- Adult and Covertypes need deeper trees (depth ≈ 12 – 16) to capture feature interactions; shallow trees underfit both.
- Rising accuracy with depth is not automatically overfitting — OOB and test curves rise together.

THE VERDICT

Accuracy Is Not Enough

Macro-F1 exposes whether minority classes are being served, not only whether most predictions are correct.

Accuracy – Macro-F1 gap (percentage points)



Source: canonical five-fold head_to_head_summary.csv / report Table III

Why this matters

Accuracy counts total correct predictions.

Macro-F1 gives each class equal voice.

Covertypes exposes the risk: AdaBoost has 58.0% accuracy but only 32.0% macro-F1.

Verdict: accuracy tells the headline; macro-F1 tells whether the headline ignores rare classes.

Is the Gap Real? Paired, Holm-Corrected Tests

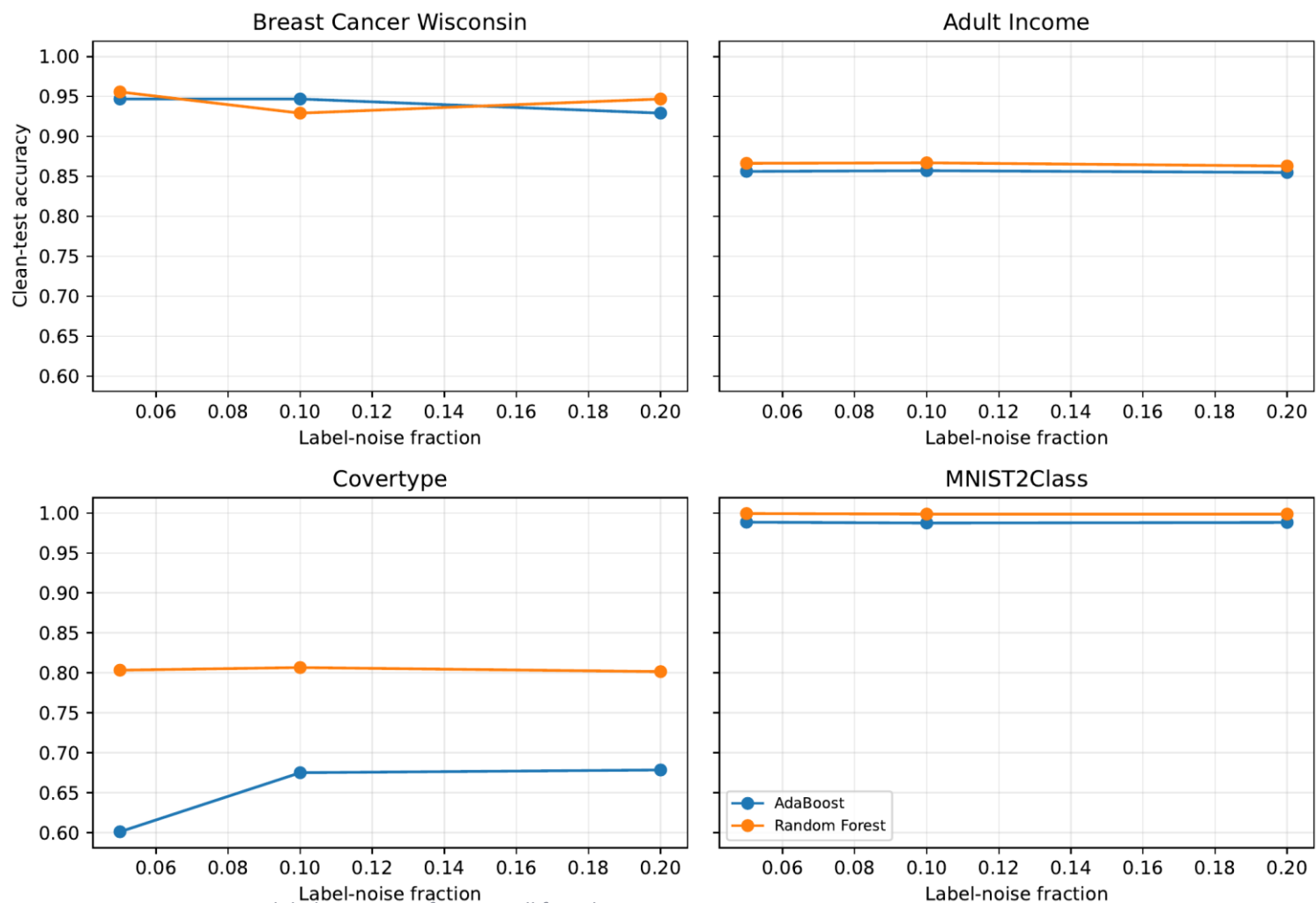
Same 5 fold indices for every model — paired t-tests isolate the model effect from fold-to-fold variation.

Dataset	Accuracy Δ (RF – Ada)	Holm-adjusted p	Verdict
Breast Cancer Wisconsin	+0.0017	0.862	Not significant
Adult Income	+0.0091	< 0.001	Significant — RF
Covertime	+0.2326	< 0.001	Significant — RF
MNIST2Class	+0.0004	0.289	Not significant

Holm-Bonferroni correction controls the family-wise error rate across all four dataset comparisons, so we don't overstate significance from running multiple tests. Adult Income and Covertime clear the corrected threshold decisively; Breast Cancer and MNIST2Class remain genuine near-ties — mean accuracy alone would have hidden this distinction.

Robustness to Training-Label Noise

Only training labels are corrupted (5% / 10% / 20%); test labels stay clean throughout.



Test accuracy vs. training-label corruption fraction, all four datasets

0.80–0.81

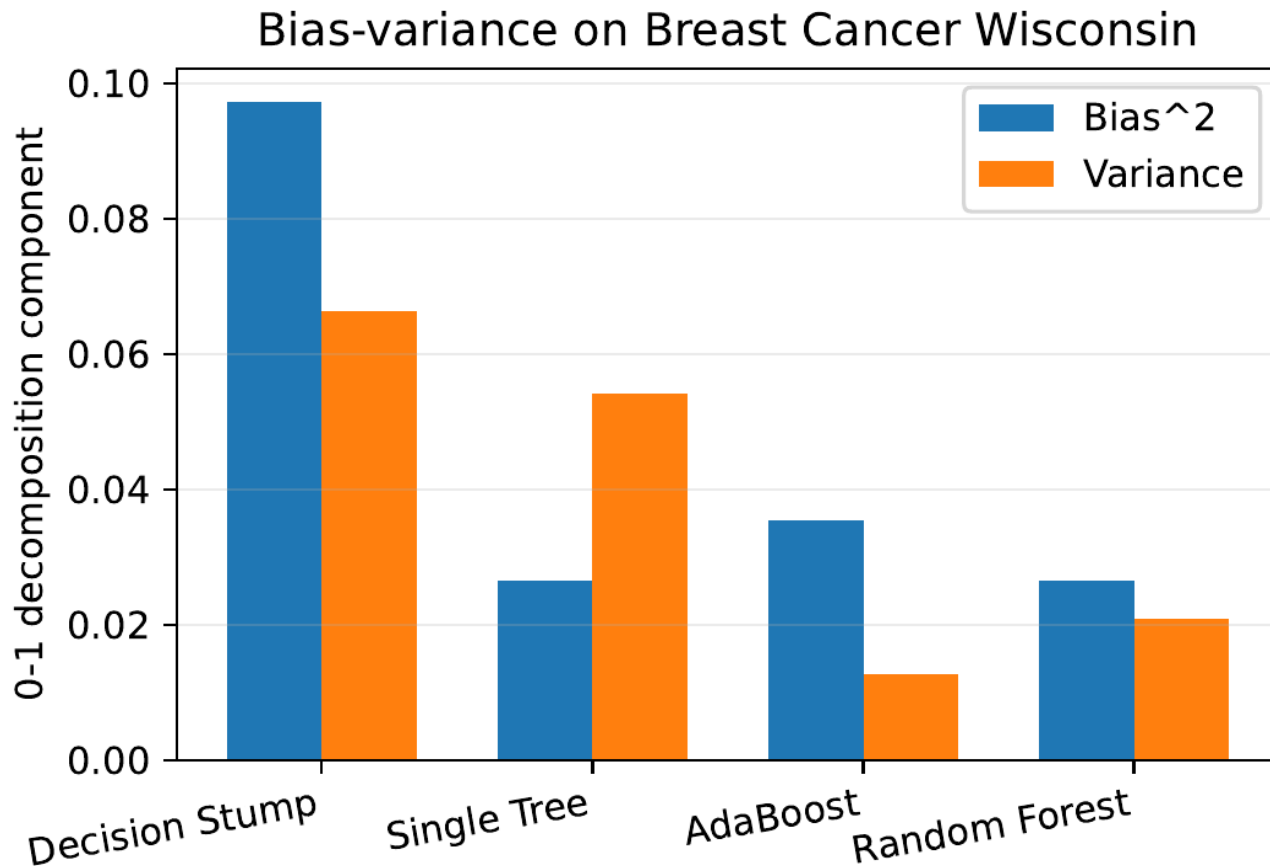
Covertypes RF accuracy range (5%–20% noise)

0.60–0.68

Covertypes AdaBoost accuracy range (5%–20% noise)

Random Forest is markedly more stable under label corruption — bootstrap averaging dilutes the influence of mislabeled training rows. This holds most visibly on Covertypes. Behavior is not strictly monotonic with noise level for either model, so we report ranges rather than claiming a clean linear trend.

Bias-Variance Decomposition (Breast Cancer, 100 Bootstraps)



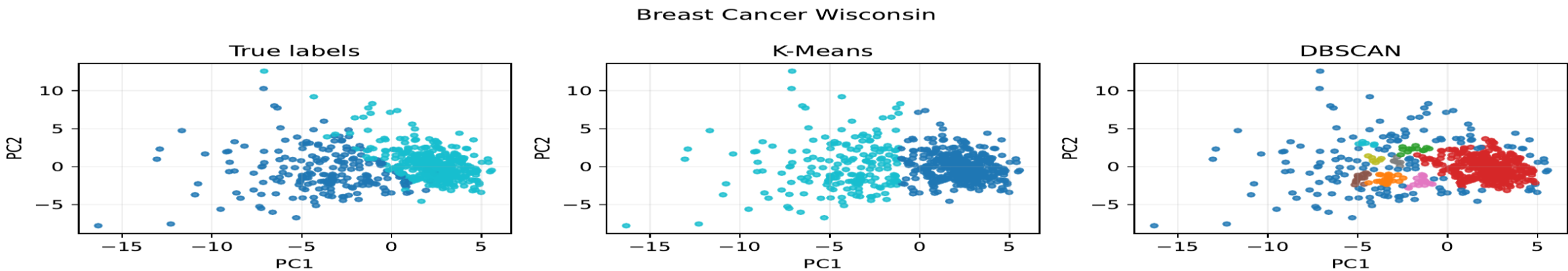
Canonical output — 100 bootstrap repetitions, Breast Cancer Wisconsin (real, not synthetic)

Model	Bias ²	Variance	Expected Loss
Decision Stump	0.097	0.066	0.114
Single Tree	0.027	0.054	0.075
AdaBoost	0.035	0.013	0.038
Random Forest	0.027	0.021	0.047

Random Forest cuts variance by 60% versus a single tree (0.054 → 0.021) while bias stays essentially unchanged (0.027 → 0.027). Bagging averages away the instability of individual trees; it does not change what the underlying model family can represent.

Structure Without Labels: PCA, K-Means, DBSCAN

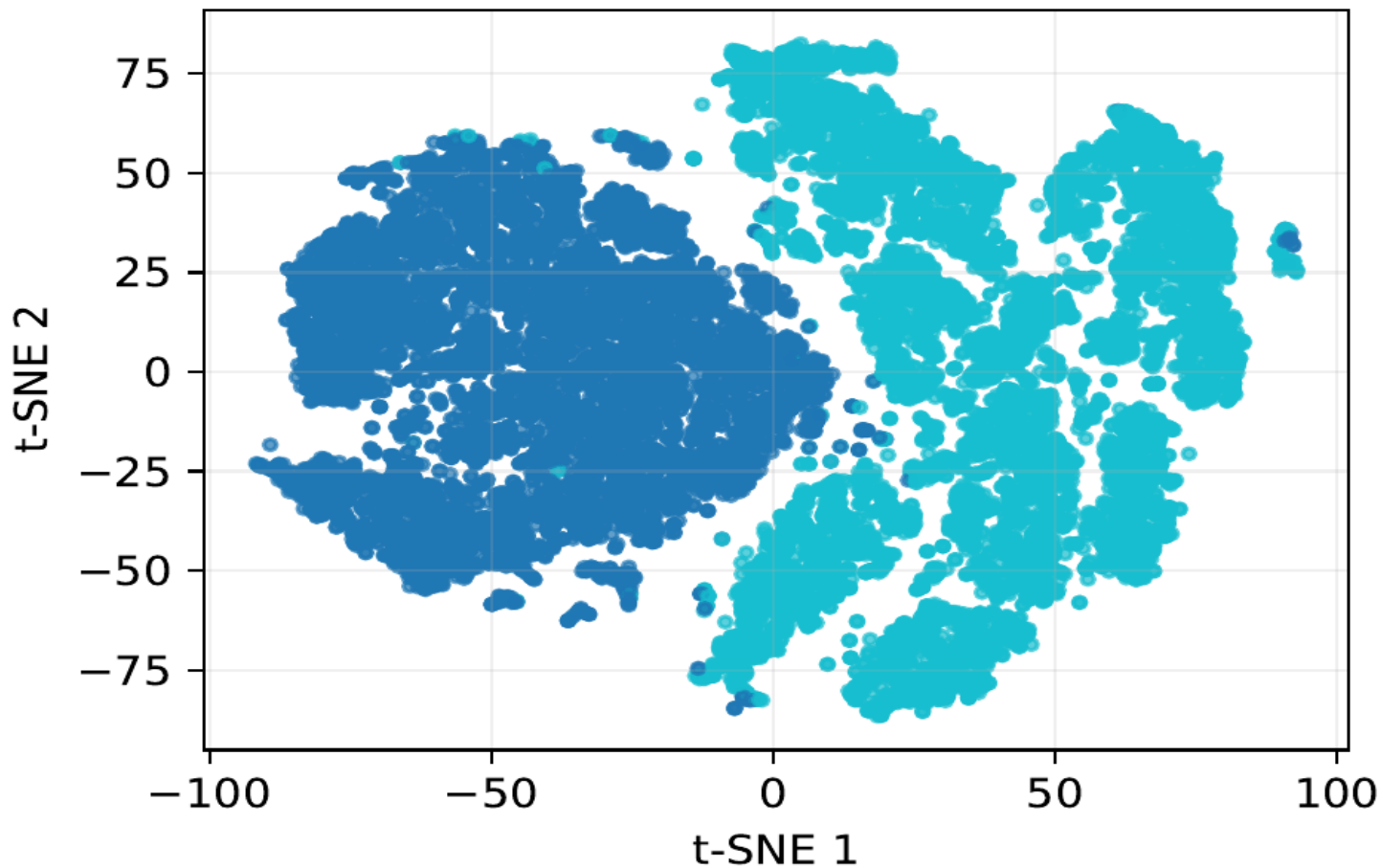
Dataset	Comp. for 90% Var.	K-Means (best k, ARI)	DBSCAN ARI
Breast Cancer Wisconsin	7	k=2, 0.659	0.450
Adult Income	22	k=3, 0.132	0.063
Covertypes	39	k=4, 0.091	0.133
MNIST2Class	150	k=2, 0.948	0.921



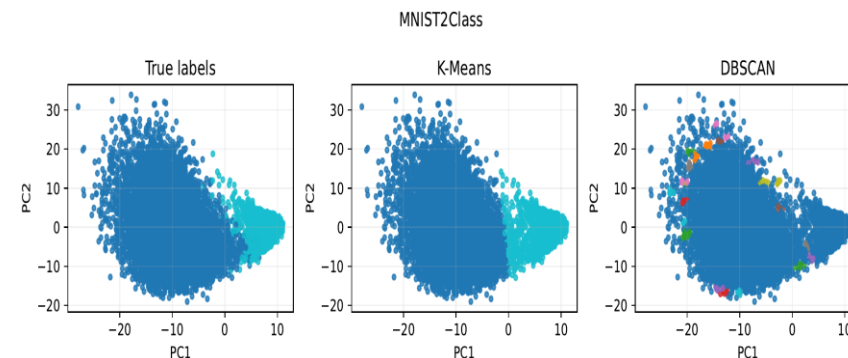
Breast Cancer Wisconsin — PCA projection colored by K-Means cluster vs. true diagnosis

MNIST2Class: High Dimensions, Clear Geometry

t-SNE bonus: MNIST2Class (all 14,780 rows)



t-SNE embedding of all 14,780 rows, colored by digit class



PCA projection, K-Means vs. true label

- 784 raw pixel features, yet 0s and 1s separate almost perfectly in 2 dimensions.
- t-SNE uses all 14,780 rows — no subsampling — to build the embedding.
- t-SNE axes carry no physical meaning; only local neighborhood structure is meaningful, not global distances.
- High nominal dimensionality does not imply a hard classification problem when class-discriminative structure is simple.

CONCLUSION

Key Findings & Limitations

- Random Forest is the safer default under imbalance, feature interactions, and heavier noise.
- Ensemble size helps only until saturation — both methods show clear diminishing returns.
- Paired, Holm-corrected significance testing separates real effects from mean-score illusions.
- AdaBoost excels when weak learners make stable, well-behaved corrections on separable data.
- Leakage-safe preprocessing discipline is a precondition for any of these numbers to be trustworthy.
- Unsupervised geometry mirrors supervised difficulty — easy-to-cluster data is easy to classify.

LIMITATIONS

Fixed hyperparameter grids and only 5 CV folds · one canonical seed (42) · t-SNE is a visualization aid, not a metric · clustering labels are descriptive, not pure unsupervised discovery · custom implementations are educational, not optimized library replacements.

The right ensemble is not a fixed choice — it is a function of the data you actually have.